

YASH YADAV (69)

Double-click (or enter) to edit

```
!pip install opencv-contrib-python scikit-image
```

```
Requirement already satisfied: opencv-contrib-python in /usr/local/lib/pyt
Requirement already satisfied: scikit-image in /usr/local/lib/python3.13/d
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.13/dist-
Requirement already satisfied: scipy>=1.11.4 in /usr/local/lib/python3.13/
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.13/
Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.13/d
Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/py
Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/pytho
Requirement already satisfied: packaging>=21 in /usr/local/lib/python3.13/
Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import exposure

print("All libraries loaded successfully!")
```

```
All libraries loaded successfully!
```

```
from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully!")
print("Image size:", img.shape)
```

yuichi-kag...nsplash.jpg

**yuichi-kageyama-4ByFHyNdoD4-unsplash.jpg**(image/jpeg) - 753073 bytes, last modified: 9/15/2026 - 100% done

Saving yuichi-kageyama-4ByFHyNdoD4-unsplash.jpg to yuichi-kageyama-4ByFHyN  
Image loaded successfully!  
Image size: (2700, 2178, 3)

```
plt.figure(figsize=(10, 6))  
plt.imshow(img_rgb)  
plt.title("Original Butterfly Image")  
plt.axis("off")  
plt.show()
```

Original Butterfly Image



```
sift = cv2.SIFT_create()  
  
keypoints, descriptors = sift.detectAndCompute(gray, None)  
  
print("Number of SIFT Keypoints:", len(keypoints))  
print("Descriptor Shape:", descriptors.shape)
```

```
Number of SIFT Keypoints: 16975  
Descriptor Shape: (16975, 128)
```

```
sift_image = cv2.drawKeypoints(  
    img_rgb,  
    keypoints,  
    None,  
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS  
)  
  
plt.figure(figsize=(12, 8))  
plt.imshow(sift_image)  
plt.title("SIFT Keypoints")
```

```
plt.axis("off")  
plt.show()
```

### SIFT Keypoints

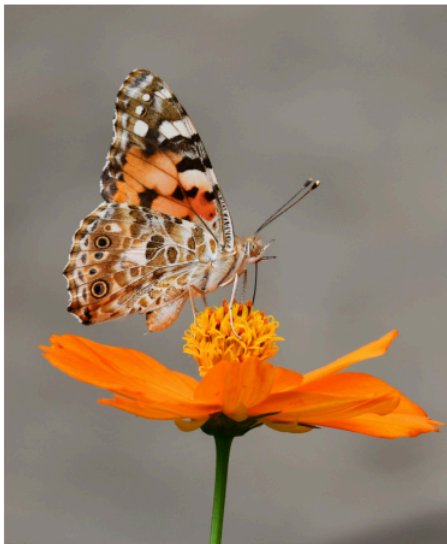


```
hog_features, hog_image = hog(  
    gray,  
    orientations=9,  
    pixels_per_cell=(8, 8),  
    cells_per_block=(2, 2),  
    block_norm='L2-Hys',  
    visualize=True  
)  
  
print("HOG Feature Extraction Completed!")  
print("HOG Feature Vector Length:", len(hog_features))
```

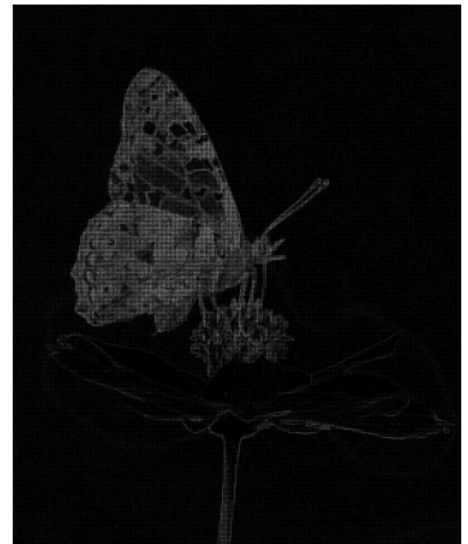
```
HOG Feature Extraction Completed!  
HOG Feature Vector Length: 3278016
```

```
hog_image_rescaled = exposure.rescale_intensity(  
    hog_image,  
    in_range=(0, 10)  
)  
  
plt.figure(figsize=(14, 6))  
  
plt.subplot(1, 2, 1)  
plt.imshow(img_rgb)  
plt.title("Original Image")  
plt.axis("off")  
  
plt.subplot(1, 2, 2)  
plt.imshow(hog_image_rescaled, cmap="gray")  
plt.title("HOG Visualization")  
plt.axis("off")  
  
plt.show()
```

Original Image



HOG Visualization



```
img2 = cv2.resize(  
    img,  
    None,  
    fx=5.2,  
    fy=5.2  
)  
  
gray2 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)  
  
print("Second image created!")
```

Second image created!

```
img2 = cv2.resize(  
    img,  
    None,  
    fx=1.2,  
    fy=1.2  
)
```

```
gray2 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)
img2_rgb = cv2.cvtColor(img2, cv2.COLOR_BGR2RGB)

print("Second image created!")
```

Second image created!

```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

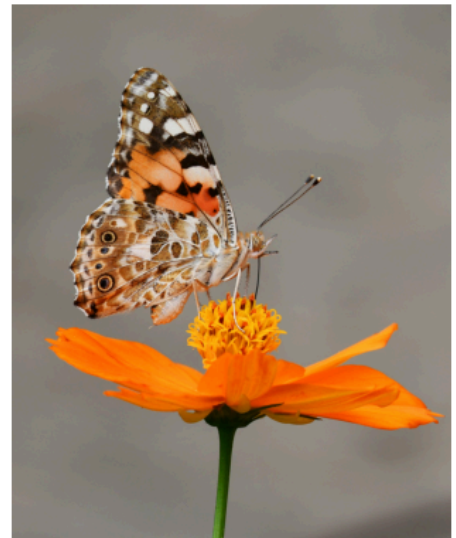
plt.subplot(1, 2, 2)
plt.imshow(img2_rgb)
plt.title("Resized Image (1.2x)")
plt.axis("off")

plt.show()
```

Original Image



Resized Image (1.2x)



```
kp1, des1 = sift.detectAndCompute(gray, None)
kp2, des2 = sift.detectAndCompute(gray2, None)

print("Original Image Keypoints:", len(kp1))
print("Second Image Keypoints:", len(kp2))
```

Original Image Keypoints: 16975  
Second Image Keypoints: 18875

```
bf = cv2.BFMatcher()

matches = bf.knnMatch(des1, des2, k=2)

good_matches = []

for m, n in matches:
```

```
        if m.distance < 0.75 * n.distance:
            good_matches.append(m)

print("Total Matches:", len(matches))
print("Good Matches:", len(good_matches))
```

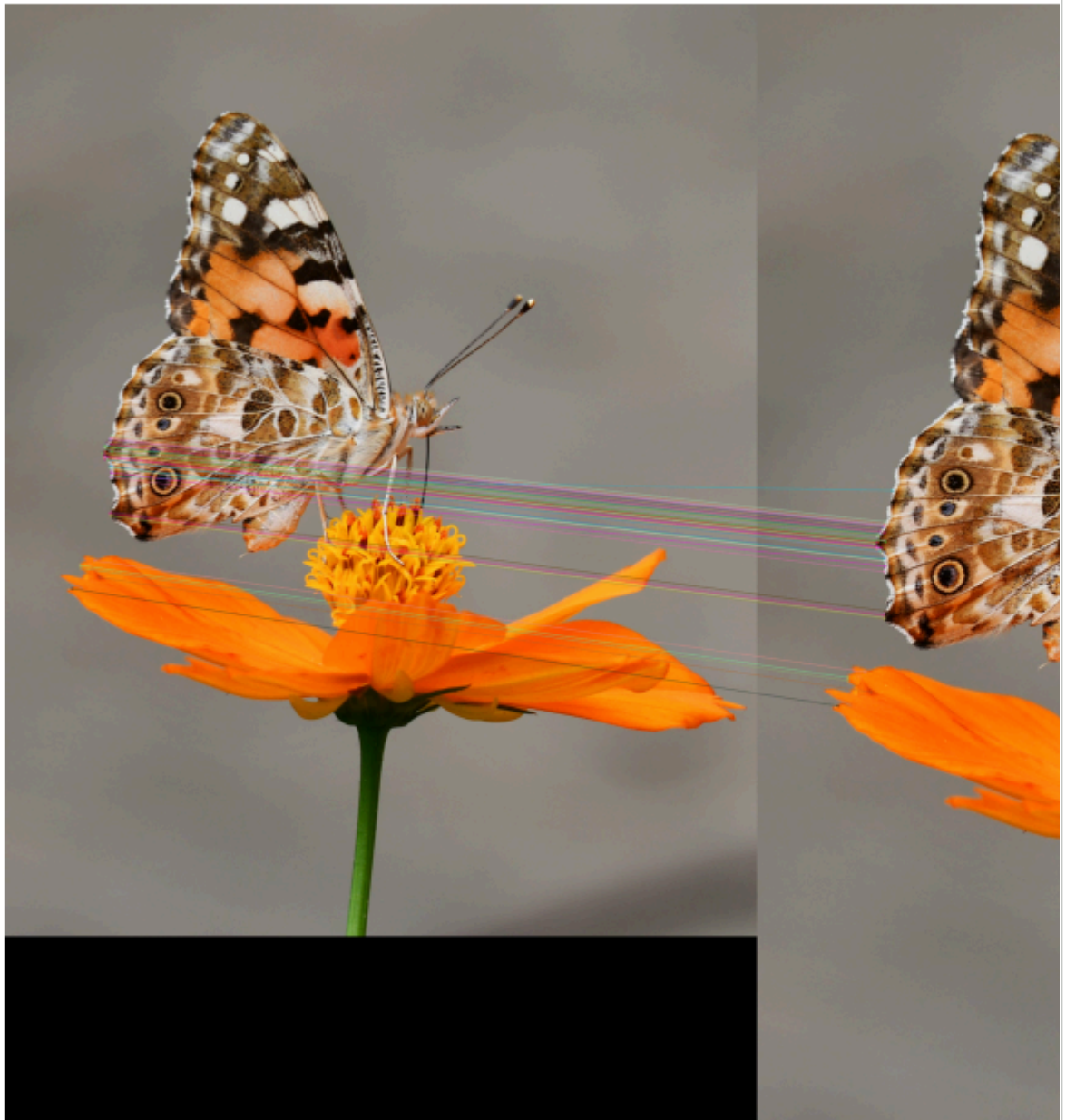
```
Total Matches: 16975
Good Matches: 11766
```

```
matched_image = cv2.drawMatches(
    img,
    kp1,
    img2,
    kp2,
    good_matches[:50],
    None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)

plt.figure(figsize=(18, 8))
plt.imshow(cv2.cvtColor(matched_image, cv2.COLOR_BGR2RGB))
plt.title("SIFT Feature Matching")
plt.axis("off")
plt.show()
```



## SIFT Feature Matching



```
plt.figure(figsize=(10, 6))  
plt.imshow(img_rgb)  
plt.title("Original Colour Image")  
plt.axis("off")  
plt.show()
```

### Original Colour Image

